Project 2 — Estimating the Truth

Building high-quality estimates from samples.

Project 2: Estimation from a simple random sample

Total points: 35 (report) + 5 (code notebook)

Course readings: Module 2: Read 0 · Read 1 · Read 2

In this project, you will use a 95% confidence interval to estimate a population proportion or population mean using data collected from a simple random sample. Complete the steps below, then submit a one-page summary and an R notebook that runs top to bottom.

Sample write-up: Download sample one-page write-up — CA high schools (PDF). Do not copy my population or variable but instead write up your own study following the grading rubric below.

Sample R notebooks (adapt to your data; choose **mean** or **proportion**):

Goal	Notebook	Open in Colab
Estimate a **population mean** with FPC	`project-2-estimation-mean-fpc.ipynb`	Open in Colab
Estimate a **population proportion** with FPC	`project-2-proportion-fpc.ipynb`	Open in Colab

After opening in Colab, use **Runtime → Change runtime type** and select **R** if needed.

Project Workflow

Phase	What
A — Milestone	Population N, frame, SRS plan, sample size n, variable type, CI method (mean vs proportion; FPC or not) — **before** full data collection
B — Full project	Collect sample, graph, check conditions, compute 95% CI, one-page summary + notebook

1. **One-page summary (PDF or equivalent)** — organized using the section headings in the rubric below.
2. **R notebook** (`.ipynb`) — all code to load your data, make figures, and compute your 95% confidence interval. The notebook must run from top to bottom.

Steps for Phase A

1. Choose a population you can list

Think of a population you could list, randomly sample from the list, and visit (or inspect) each sampled item to collect one variable, either a number or a binary outcome per item.

It is hard to randomly sample people without university research approval and without a complete contact list for the population. For these reasons, I recommend a population of non-human objects. For example:

- Number all parking spaces in a lot (e.g. from a Google Maps image), randomly select 10 spaces, and record one variable at each space (occupied or not? truck or not? white or not? model year?).
- Sample from a numbered list of items on a website, as in the sample project. All sneakers on Amazon or census tracts in California are also good lists.

Your turn.

My population is ____, which has $N =$ ________ objects.

2. Choose the parameter and variable

Once you have a population, decide what **mean** or **proportion** you want to estimate. Identify the **variable** you will collect for each sampled object.

Your turn.

My variable is ______________________________, which is binary categorical numerical (circle one), so I'll be estimating a proportion mean (circle one).

3. Choose a sample size

Larger samples give more precise estimates, but collecting data takes time. In professional work, a sample size calculation balances time and precision. For this project, collect data from 10 objects or the largest number you can feasibly visit in 1–2 hours.

Your turn.

I think the largest number of objects in my population I could feasibly visit and collect information from is $n =$ ________.

4. Draw a simple random sample

Use a **random number generator** to select n distinct items from 1 to N.

Rossman Chance random number generator or

`sample(1:50, 6)` in R samples 6 objects from the integers 1 through 50.

Your turn.

The n items I need to visit are: ____(list their numbers)

5. Collect your data

Visit each sampled item and record your variable. Keep your dataset in R (a vector or a small data frame).

Your turn.

The dataset from my sample is: ____(list value of the variable for each item)

6. Compute your point estimate

Compute the sample mean (numerical variable) or sample proportion (binary variable). This is your single best estimate of the population parameter.

Your turn.

My single best estimate of the population is ______________________

7. Choose and compute a 95% confidence interval in R

Use the Project 2 decision table and conditions checklist from Read 2.

Record and **check the conditions** for your chosen interval, then compute it in R. Start from the **mean notebook** or **proportion notebook** if helpful.

Your turn.

My 95% confidence interval estimate is $L =$ ________ and $U =$ ________.

The conditions for this interval are (sampling, model, FPC if applicable):

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8. Interpret the interval

Write one sentence using the interpretation recipe from Read 2:

> I am 95% confident that the ______________________ (parameter: mean or proportion) of ______________________ (variable) for ______________________ (population) is between ________ (L) and ________ (U).

9. Write your one-page summary

Add an appropriate plot of your data to your code notebook (histogram or dotplot for numerical variable, bar plot for binary). Then follow the grading rubric below for the report.

Grading rubric for one-page summary

Total points: 35

1. Introduction (3 pts)

- Briefly describe in a single sentence the parameter (mean or proportion) you are estimating.
- Clearly state the population associated with this parameter.
- Briefly describe your motivation for choosing to estimate this quantity.

2. Methods

2a. Data collection (8 pts)

- Include a concise description of your population, including its size N.
- Describe how you obtained a simple random sample from a list of the population (e.g. random number generator and population size).
- Describe exactly how you located an item corresponding to each random number (e.g. "I set the number of schools shown on a page to 100…").
- Describe the variable you collected from each sampled item. State clearly whether you collected a number or a binary outcome. If numerical, include units.

2b. Data analysis (6 pts)

- Describe how you computed the 95% confidence interval in R (e.g. `survey`, `t.test`, `prop.test`, or `binom.test`).
- State whether you used the FPC and why.
- Describe the conditions checklist for your interval: SRS (+ BRP if proportion), large-sample / model rule, and FPC decision — and whether you believe each is met.

3. Results

3a. Descriptive statistics (6 pts)

- Include a well-labelled graph of your dataset.
- Include the mean and SD if the variable is numerical, or the count and proportion if the variable is binary

categorical.

3b. Inferential statistics (6 pts)

- Include the values of L and U from your 95% confidence interval.
- Include a one-sentence interpretation of the interval in the context of your parameter and population.

4. Conclusion (6 pts)

- Briefly describe your reaction to the final estimate — is it about what you expected?
- Briefly state whether you believe your estimate is biased (center of estimates vs. parameter — was SRS from a good frame?).
- Briefly state how you could make your estimate more precise (spread / CI width — e.g. larger n after fixing design).